

TraceFlow 1.0.0: Validation Pack

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Risk Register

TraceFlow libraries track software hazards alongside the requirements and tests that mitigate them.

Risk Scoring Method

Each risk is assessed on two dimensions, **Severity** and **Probability**. Each dimension is scored on the scale below, and the overall risk rating is the product of the two scores (Severity × Probability).

Level	Score
Low / Minor / Rare / Unlikely	1
Medium / Moderate / Possible	3
High / Major / Frequent	4
Very High / Critical / Catastrophic	5

The resulting product determines the overall risk band:

Band	Score range	Colour
Critical	16+	
High	9–15	
Medium	4–8	
Low	1–3	

The matrix below shows every combination of Severity and Probability, the resulting score, and the band into which that score falls.

Severity	Probability			
	Low (1)	Medium (3)	High (4)	Very High (5)
Very High (5)	5 (Medium)	15 (High)	20 (Critical)	25 (Critical)
High (4)	4 (Medium)	12 (High)	16 (Critical)	20 (Critical)
Medium (3)	3 (Low)	9 (High)	12 (High)	15 (High)
Low (1)	1 (Low)	3 (Low)	4 (Medium)	5 (Medium)

Risk Register

Risk ID	Hazardous Situation	Harm	Cause	Risk (Severity × Probability)	Controls	Residual Risk
RISK-001: Incorrect study-patient association	Clinician views the wrong patient's images believing they are correct.	Misdiagnosis or inappropriate treatment.	Race condition during HL7/DICOM message processing leading to incorrect PatientID or AccessionNumber.	High × Medium = High (12)	Identifier reconciliation service (REQ-005) exercised by TEST-002.	Medium × Low = Low (3) Operator review plus automatic quarantine when mismatches occur.
RISK-002: Undetected imaging artifacts	Poor quality MRI acquisitions flow through the analysis pipeline without alerts.	False clinical conclusions or delayed treatment.	Lack of automated QA gates between the import module (REQ-002) and analysis pipeline (REQ-003).	Medium × Medium = High (9)	Automated test matrix (TEST-003) and image QC hooks inside the pipeline.	Low × Low = Low (1) Clinician-facing dashboard highlights residual warnings for manual acknowledgement.
RISK-003: Audit trail corruption	Investigators cannot reconstruct prior runs because audit entries were dropped.	Compliance breach or inability to support safety investigations.	Misconfigured storage or missing webhook callbacks when persisting audit artifacts from REQ-005.	Medium × Low = Low (3)	Continuous integration checks (TEST-003) and deployment checklist in docs/ius.md.	Low × Low = Low (1) Acceptable provided long-term storage health checks pass weekly.
RISK-004: Unauthorized workspace access	An unapproved clinician launches the UI and edits live studies without authentication.	Breach of patient confidentiality and risk of tampering with diagnostic evidence.	Misconfigured identity provider or bypass of the login screen covered in REQ-001.	High × Low = Medium (4)	OpenID Connect gateway, MFA, and TEST-001 regression runs.	Medium × Low = Low (3) Acceptable when the security team performs quarterly access recertification.
RISK-005: Notification backlog for identifier mismatches	Identifier reconciliation alerts pile up, so clinicians miss urgent mismatches.	Delayed detection of study/patient mix-ups (see REQ-005).	Downstream webhook queue saturation or operator console outages.	Medium × Medium = High (9)	Autoscaling webhook workers plus the workflow exercised by TEST-002.	Low × Medium = Low (3) Escalate to the on-call operator when more than five alerts remain untriaged for over 30 minutes.
RISK-006: Exported report mismatch	Generated PDF or DICOM SR exports omit the source dataset IDs.	Inability to justify clinical findings or to reconstruct the original evidence trail.	Faulty serialization of the output schema defined in REQ-004.	Medium × Low = Low (3)	Automated CI pipeline (TEST-003) that validates exports against golden files.	Low × Low = Low (1) Acceptable when QA signs off release bundles that include checksum manifests.
RISK-007: Pipeline drift between releases	Model weights or preprocessing steps change without validation, yielding inconsistent diagnoses.	False positives/negatives that could trigger incorrect therapy.	Manual edits to REQ-003 pipeline components without rerunning validation.	High × Medium = High (12)	Continuous integration gates in TEST-003 plus code reviews for every pipeline update.	Medium × Low = Low (3) Acceptable once deployment reports attach the validation pack and a signed change record.
RISK-008: Data retention policy breach	Backups omit audit logs, preventing reconstruction of historic identifier fixes.	Compliance violations with ICH-GCP and inability to support investigations.	Retention scripts ignore the storage bucket defined in REQ-005.	Medium × Medium = High (9)	Scheduled integrity checks plus TEST-003 to validate log export scripts.	Low × Low = Low (1) Acceptable when quarterly SOP reviews confirm retention evidence is archived.
RISK-009: Test evidence missing	Validation evidence referenced by TEST-003 is not included in the validation pack.	Release approvals rely on incomplete evidence trails.	Operator does not attach the checklist evidence for manual test execution.	Medium × Medium = High (9)	Release checklist gates in docs/ius.md require evidence attachment before sign-off.	Low × Low = Low (1) Acceptable provided release managers verify example.pdf before sign-off.
RISK-010: External interface misconfiguration	Downstream systems consume reconciliation webhooks with the wrong schema and silently drop events.	Identifier mismatches never reach clinical users even though TraceFlow generated the alerts.	Integration guides diverge from the specification in REQ-005 or design doc steps.	Medium × Medium = High (9)	Installation & User Specification walkthrough (docs/ius.md) and contract tests executed as part of TEST-002.	Low × Medium = Low (3) Acceptable once each release captures signed integration test outputs in the validation pack.
RISK-011: Manual test paperwork incomplete	Manual acceptance tests (TEST-001) are executed but signatures or pass/fail states are unsigned, invalidating the safety case.	Regulatory submissions may be rejected, delaying patient access.	Clinicians forget to fill the interactive PDF fields exported by TraceFlow.	Low × Medium = Low (3)	The test cover sheet plus electronic signature workflow baked into TraceFlow forms.	Low × Low = Low (1) Acceptable when QA audits confirm PDF form completion before release.